

ITU-MiniTwit - Group e - Report

Frederik Hørup , Marie Johansen , Nikolej Lundquist , Sara Bagger , Vitus Brodersen <>

May 12, 2026

Contents

1	Introduction	1
2	System	1
2.1	Architecture and Design	1
2.2	Dependencies	1
2.3	System States	1
3	Process	2
3.1	CI/CD	2
3.2	Monitoring	2
3.3	Logging	2
3.4	Security	2
3.5	Availability and Scaling	2
4	Reflection	2
5	Use of Generative AI	2

1 Introduction

Authors:

2 System

2.1 Architecture and Design

Authors:

2.2 Dependencies

Authors:

2.3 System States

Authors:

3 Process

3.1 CI/CD

Authors: Vitus

Continuous integration and deployment for this project happens in a semi-automatic set of parallel steps. When a developer creates a pull request on the github repository, multiple actions and third-party tools begin checking the validity of the request. Below are the tools:

- Static analysis and security github action Is a compound action for linting, formatting, and security:
 - Hadolint: Dockerfile Linter
 - CSHarrier: C# Formatter
 - Roslyn: C# Linter
 - CodeQL: Static Code Security
 - Docker Scout: Docker Image Vulnerability scanner
- Codacy static analysis
 - asd
- SonarCloud code analysis
 - asd
- Build and Test
 - asd
- Staging and Deployment
 - asd

3.2 Monitoring

Authors:

3.3 Logging

Authors:

3.4 Security

Authors: Sara - Reverse proxy, https using tls certif - Firewall - In CI/CD: security analysis tools - Docker hardened images: switching services to DHI - Grafana as non-root user?

3.5 Availability and Scaling

Authors:

4 Reflection

Authors:

5 Use of Generative AI

Authors: Nikolej

During the development of this project we used ChatGPT in two main ways:

Debugging. Often when encountering unexpected bugs and error messages, we would consult ChatGPT about the cause and potential fixes While it could rarely fix the issues entirely by itself, more often than not, it pointed us in the right direction.

Research. This course presents challenges involving numerous technologies, most of which we were unfamiliar with. As such, ChatGPT was used to summarize lengthy documentation and provide concrete guides tailored to our situation.

Generating boilerplate / configurations. ChatGPT was also used for simple tasks like generating boilerplate code or writing Github Actions Workflow files. For example, the `build-report.yml` workflow, that converts the markdown source into a pdf.

The use of AI has made these tasks easier, spared us many frustrations and saved considerable time during development. On the flip side, AI may have deprived us the extensive knowledge and experience you gain by, for example, painstaking reading of documentation or spending hours resolving a tiny bug.